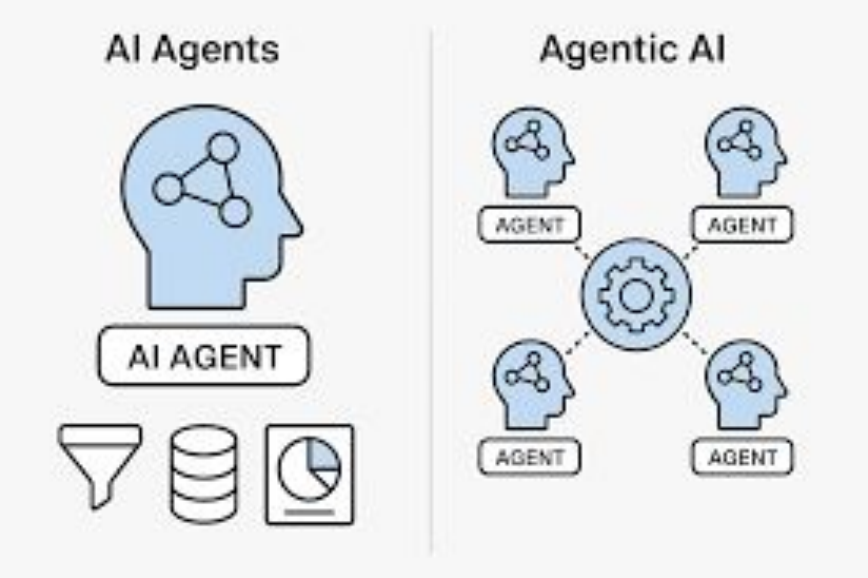
OpenClaw

By Matthew Hawkes and Matteo Bloom

What is Agentic AI?

Core Capabilities:

- Perception: Analyze logs, APIs, and databases to understand their environment.
- Reasoning & Planning: Break down high-level objectives into multi-step plans.
- Tool Use: They interact with external software.
- Memory: They maintain context, recognizing patterns from prior interactions



Agenda

01 Architecture Overview

02 Training Protocols

03 Key Innovations

04 Common Tasks

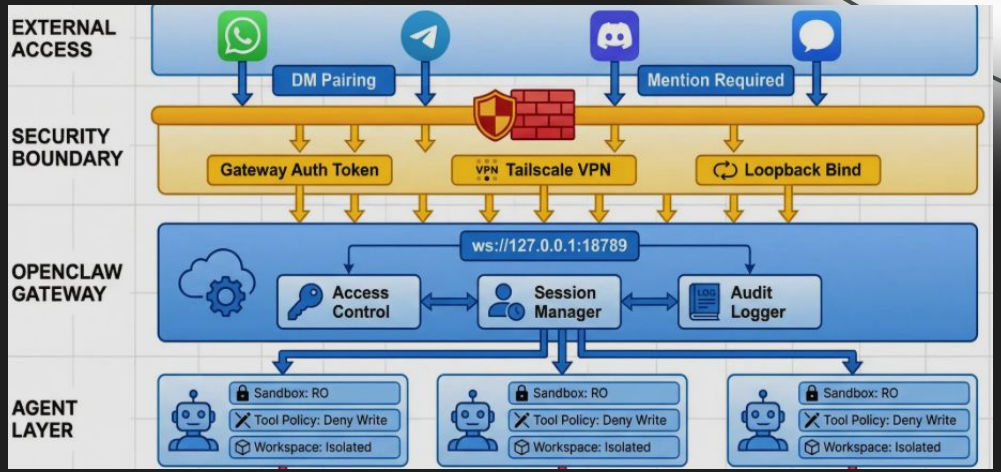
05 Real-World Applications

06 Shortcomings

Architecture Overview

What OpenClaw Is Built On

- Transformer-based architecture
- Multi-head self-attention for contextual reasoning
- Token + positional embeddings
- Deep stack of transformer layers
- Layer normalization + residual connections for stability



In-depth Architecture

LLM Reasoning Core

- Powered by transformer-based models
- Handles NLP, chain-of-thought reasoning, and decision making.
- Breaks tasks into steps
- OpenClaw's "brain"

Tool Use Layer

- Allows the AI to go beyond a conversation
- Examples:
  - Fill out forms, send messages, execute code, download software

Memory System

- Short-term memory
  - Conversation history
  - Intermediate reasoning
- Long-term memory
  - Vector databases

Agent Loop

1. Think
2. Act
3. Observe
4. Repeat

Enables autonomous tool use and iterative problem solving

Training Protocols

Training Data Sources

- Curated web text
- Books and long-form documents
- Technical manuals
- Code repositories
- Domain-specific corpora (legal, medical, financial)

Training Objectives

- Next-token prediction
- Instruction tuning
- RLHF (reinforcement learning from human feedback)

Optimization

- AdamW optimizer
- Learning-rate warmup + cosine decay
- Mixed-precision training for efficiency

Key Innovations

OpenClaw

'HE AI THAT ACTUALLY DOES THINGS

Efficient Attention Scaling

What it is: A streamlined attention mechanism that reduces the quadratic cost of standard transformers.

Why it matters: Faster inference and better long-context handling.

How OpenClaw uses it: Sparse attention + token pruning to keep only the most relevant context

Modular Fine-Tuning (LoRA)

What it is: Small adapter layers added to the model for cheap fine-tuning.

Why it matters: Lets teams customize OpenClaw without retraining the whole model.

How OpenClaw uses it: Plug-in LoRA modules for legal, medical, enterprise, or multi-domain tasks.

Alignment & Safety Layer

What it is: A post-training pipeline combining RLHF and rule-based filtering.

Why it matters: Reduces harmful, biased, or unpredictable outputs.

How OpenClaw uses it: Safety classifiers + human-preference tuning to guide responses.

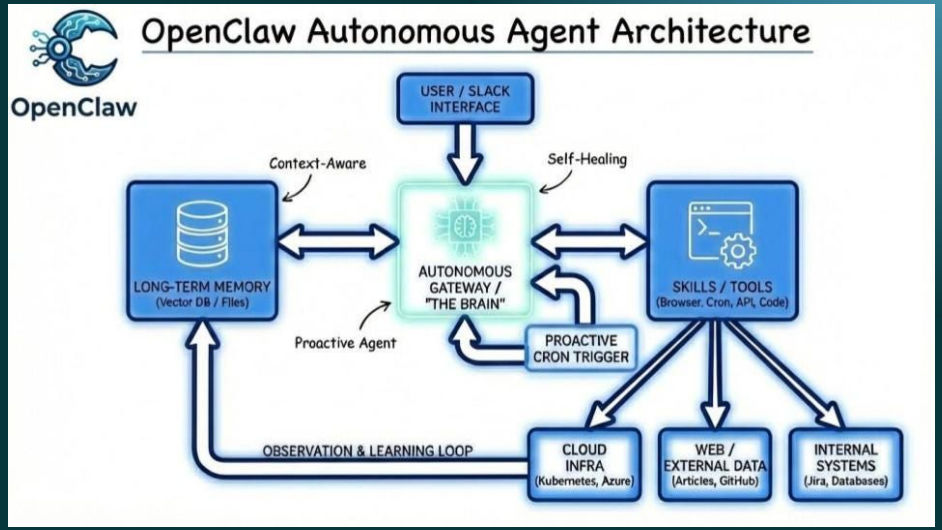
Multimodal Extensions

What it is: An optional vision encoder that lets the model process images.

Why it matters: Enables tasks like visual Q&A, captioning, and document understanding.

How OpenClaw uses it: CLIP-style encoder + cross-attention to merge image and text embeddings.

Common Tasks



Text Generation →

- Essays, explanations, creative writing
- Summaries and paraphrasing

Classification →

- Sentiment analysis
- Topic detection
- Intent classification

Knowledge Tasks →

- Domain-specific Q&A
- Retrieval-augmented reasoning

Code Tasks →

- Python, JavaScript, SQL generation
- Debugging and code explanation

Data Transformation →

- Converts messy text into structured formats
- Extracts entities (names, dates, locations, products)
- Builds tables, bullet lists, summaries, or outlines
- Useful for research, documentation, and workflow automation

Real-World Applications

Business

- Automates customer support with consistent, fast responses.
- Drafts emails, reports, proposals, and internal documentation.
- Summarizes long documents for decision-makers
- Assists with research, competitive analysis, and data extraction

Education

- Generates study guides, explanations, and practice questions
- Supports language learning and reading comprehension
- Helps students break down complex topics into simpler steps
- Assists teachers with lesson planning and content creation

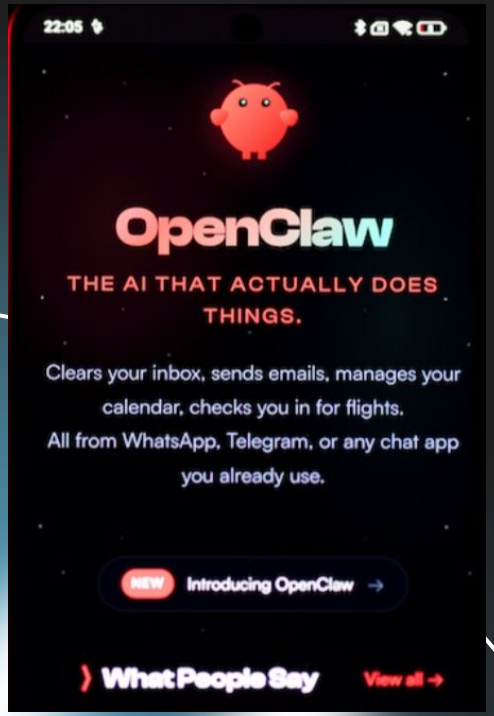
Tech

- Writes and explains code in multiple languages
- Debugs errors and suggests fixes
- Generates API documentation and technical summaries
- Helps engineers reason through system design or architecture questions

Creativity

- Brainstorms ideas for stories, scripts, and campaigns
- Produces outlines, character concepts, and dialogue drafts
- Assists with content calendars and social media planning
- Helps creators iterate quickly on early-stage concepts

Shortcomings



Bias

Fairness Issues:
- Inherits patterns from training data
- May reinforce stereotypes

Hallucinations

- Generates confident but incorrect answers
- Requires human oversight

Resource Cost/Limitations/Risks

- Expensive to train from scratch
- Requires GPUs for large-scale inference
- Long-context reasoning degrades over very long sequences
- Misinformation
- Automation misuse
- Privacy concerns

Thank you!

Questions?